

Inverse Functions

One-to-One Functions

One-to-One Function: A function that has no repeated outputs. In other words, each output value can correspond to only one input value.

In order for a relation to qualify as a function, for every input there must be only one output.

In order for a function to be one-to-one, for every output there can be only one input.

Every input has only one output and every output has only one input.

Not Functions

X	10	7	-2	7	4
Y	11	10	13	-8	3

X	0	0	83	40	-3
Y	1	2	81	46	-3

Not One-to-One Functions

X	3	2	-4	-10	9
f(x)	7	8	8	1	12

X	-3/8	3	0	11	10
f(x)	2	5	9	6	2

One-to-One Functions

X	17	10	0	-1	-6
f(x)	5	4	0	-2	-8

X	-15	20	17	36	18
f(x)	3	4	21	40	25

Relations are written below in table form. Some of the tables could represent functions that are one-to-one, some of them could represent functions that are not one-to-one, and some cannot represent functions. For each table, write "1-to-1 function," "function," or "not a function."

• ①

X	16	20	4	10	10
Y	19	32	5	32	3

Not a Function

②

X	-3	2	8	0	-2
Y	-3	8	10	0	-2

1-to-1 Function

③

X	32	33	34	35	36
Y	31	30	31	32	0

Function

④

X	-7	10	17	20	-7
Y	4	3	14	21	3

Not a Function

⑤

X	11	4	5	6	18
Y	3	9	10	11	12

1-to-1 Function

⑦

X	-1	0	3	-4	0
Y	-1	0	1	-3	5

Not a Function

• ⑨

X	20	19	18	5	3
Y	-4	-4	7	4	2

Function

⑪

X	3	-1	86	22	1
Y	7	4	85	21	1

1-to-1 Function

⑬

X	0.1	2.3	4.7	0.9	3.6
Y	20	20	20	20	19

Function

⑮

X	-2	-8	4	-3	0
Y	8	-8	2	1	7

1-to-1 Function

⑥

X	20	10	-2	-1	0
Y	-3	-5	-5	-5	-1

Function

⑧

X	10	2	1	0	3
Y	-10	-9	-8	4	2

1-to-1 Function

⑩

X	-9	10	-8	-9	17
Y	10	14	20	3	-1

Not a Function

⑫

X	-3	-1	0	3	1
Y	-2	-7	-7	-8	-7

Function

⑭

X	16	14	13	12	11
Y	15	12	11	10	9

1-to-1 Function

⑯

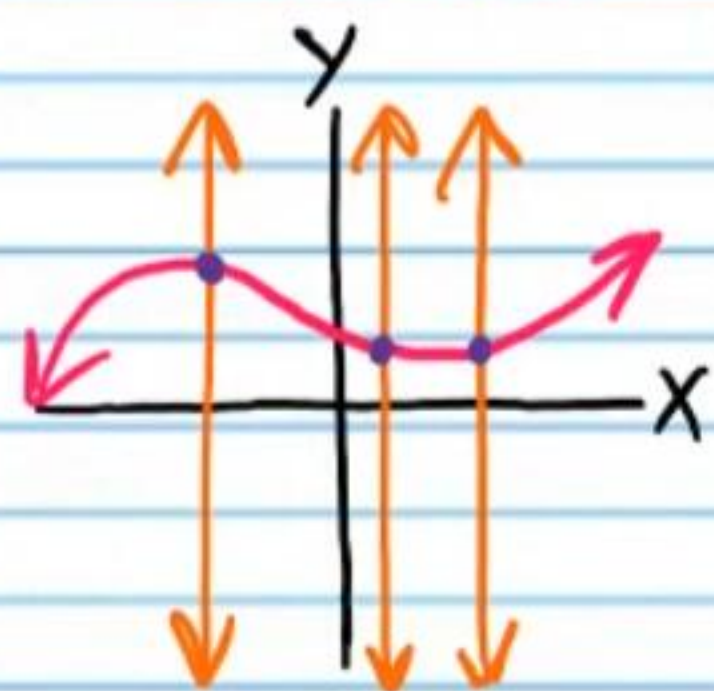
X	1	-1	1	-1	1
Y	32	42	82	72	12

Not a Function

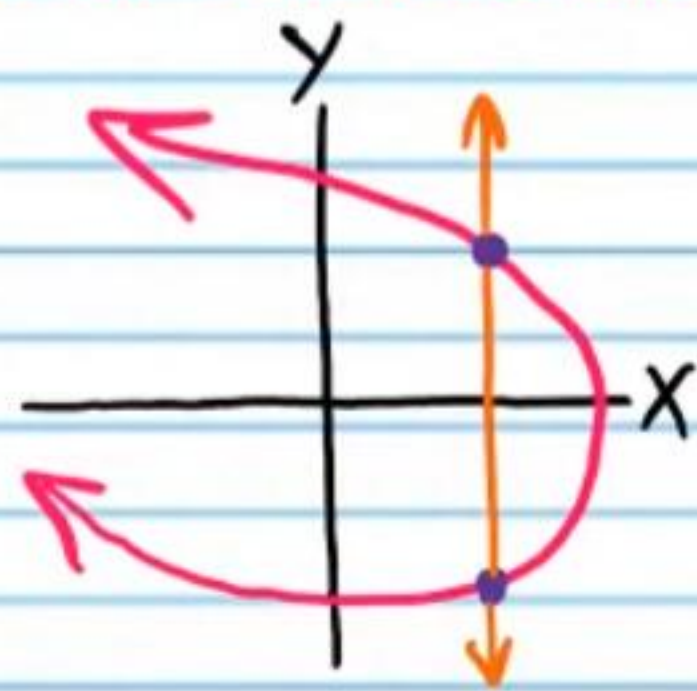
To determine if a graph can represent a function, use the Vertical Line Test.

Vertical Line Test: If any vertical line touches a graph more than once, then the graph cannot represent a function.

Function



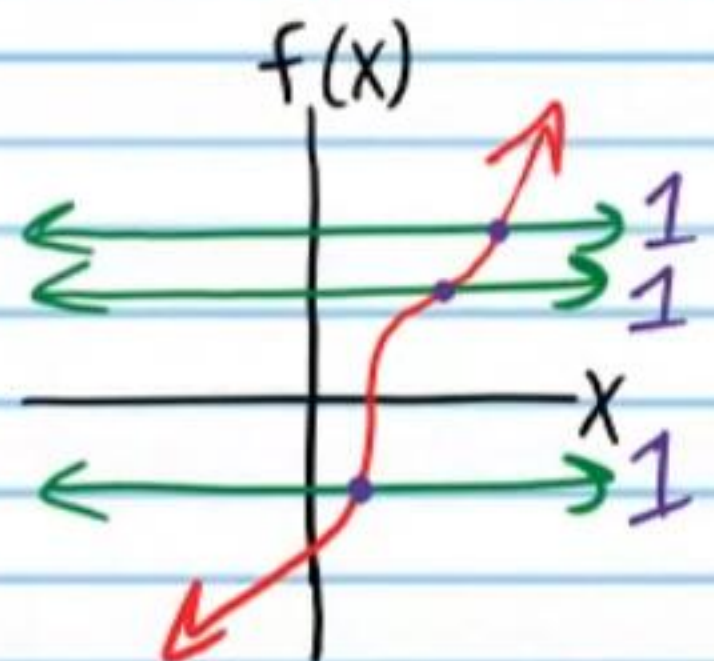
Not a Function



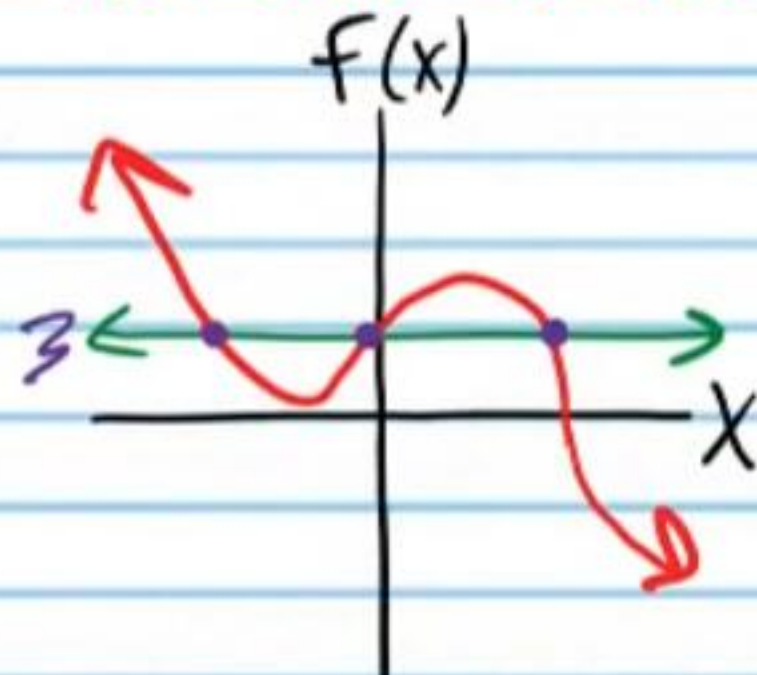
To determine if a graph represents a one-to-one function, perform the Horizontal Line Test.

Horizontal Line Test: If any horizontal line touches the graph of a function more than once, then the function is **not** one-to-one.

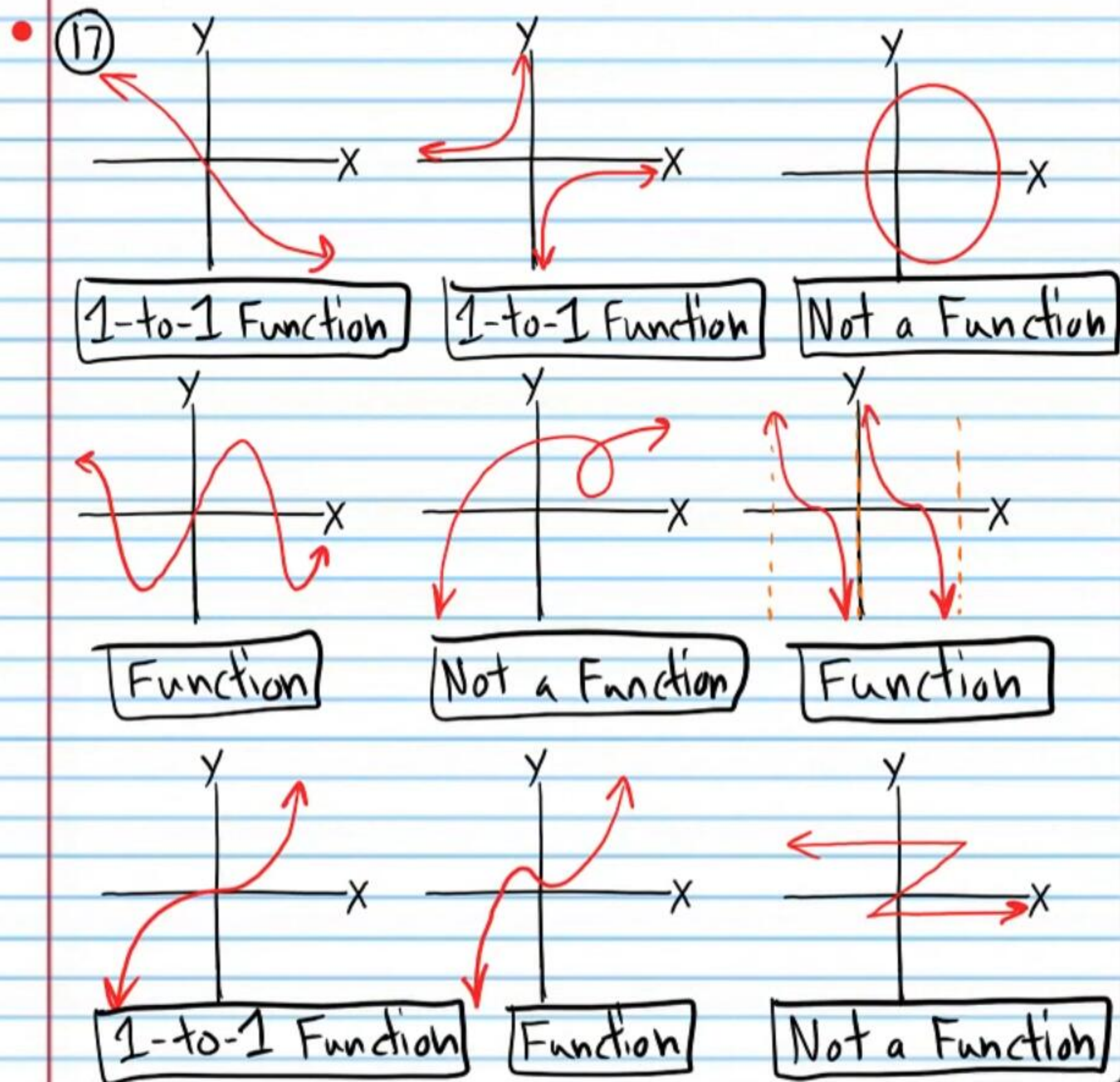
One-to-One Function



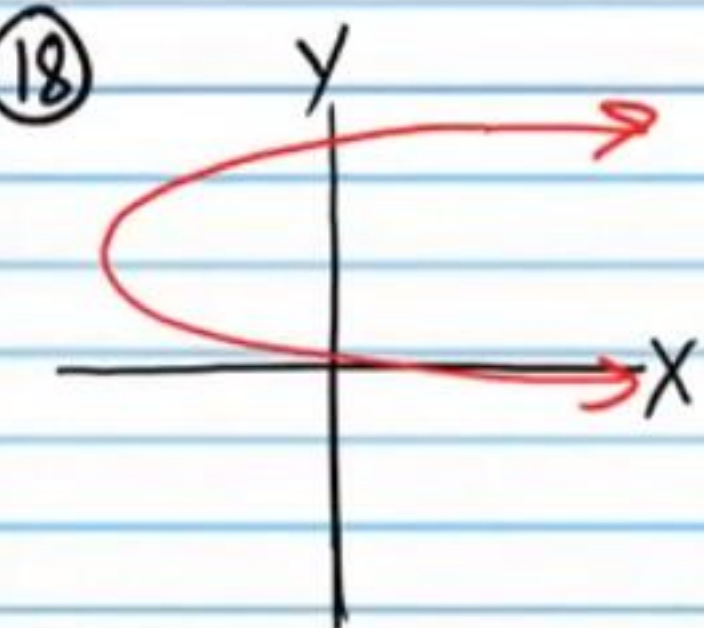
Not One-to-One



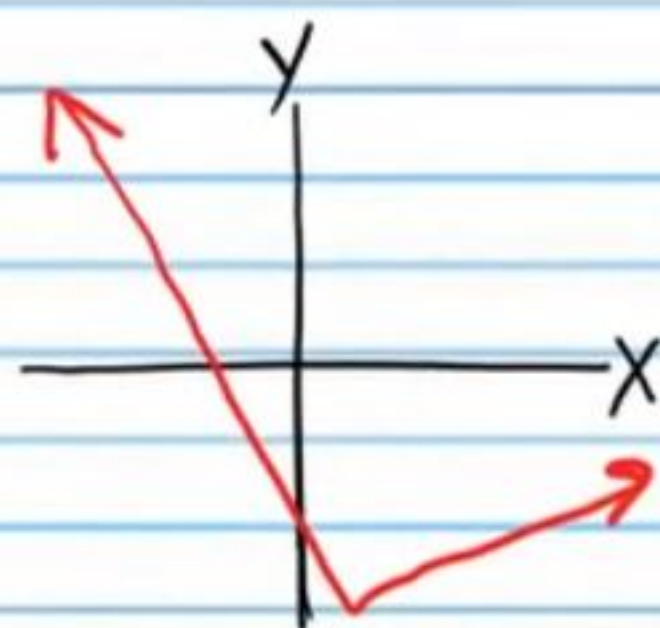
Relations are written below in the form of graphs. Some of the graphs could represent functions that are one-to-one, some of them could represent functions that are not one-to-one, and some cannot represent functions. Use the Vertical Line Test and the Horizontal Line Test to determine the scenario for each graph. Write "1-to-1 function," "function," or "not a function."



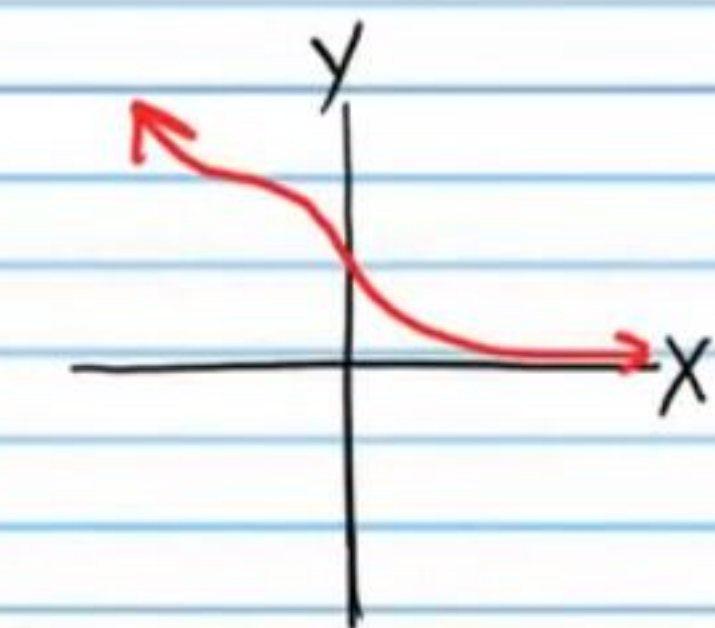
(18)



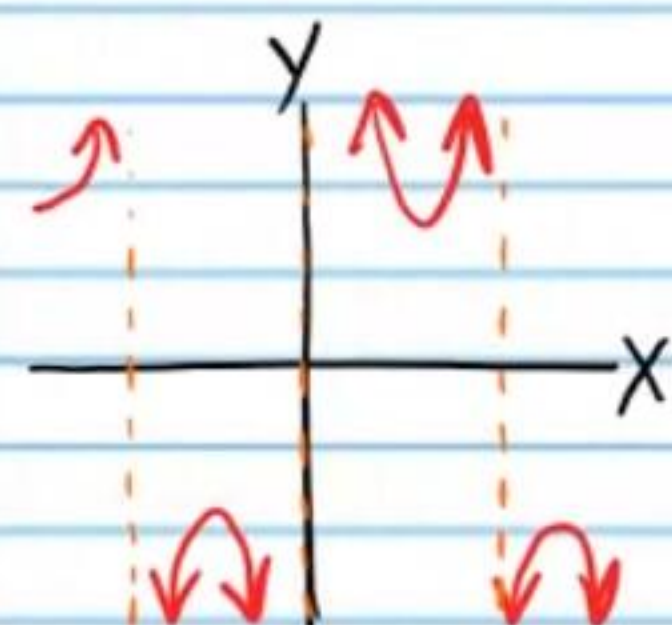
Not a Function



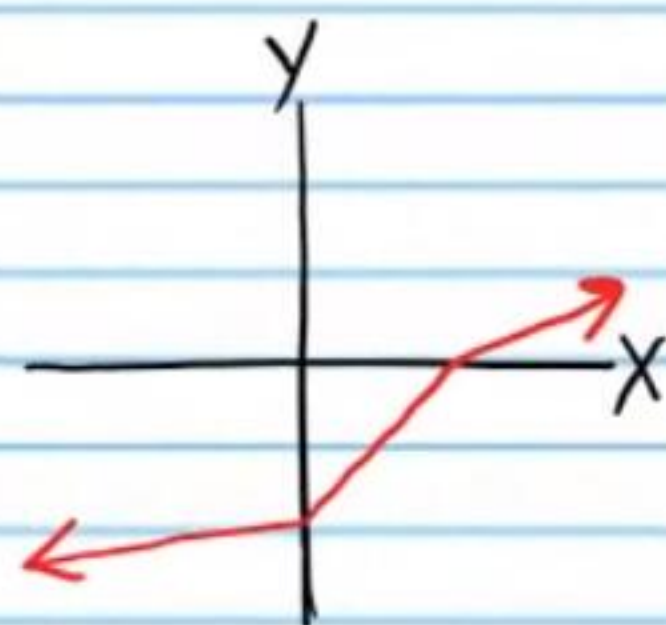
Function



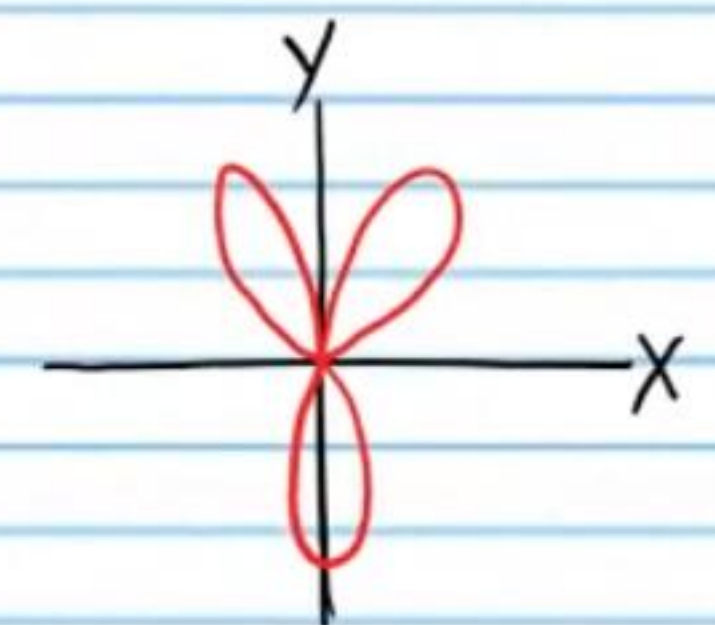
1-to-1 Function



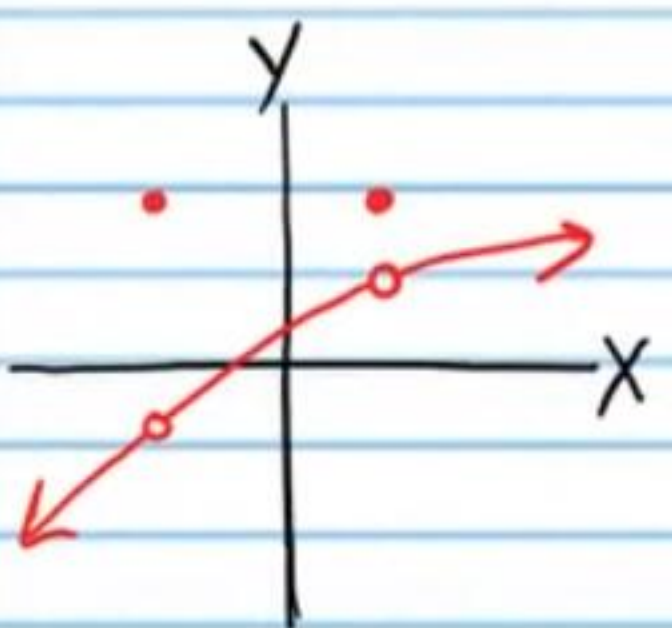
Function



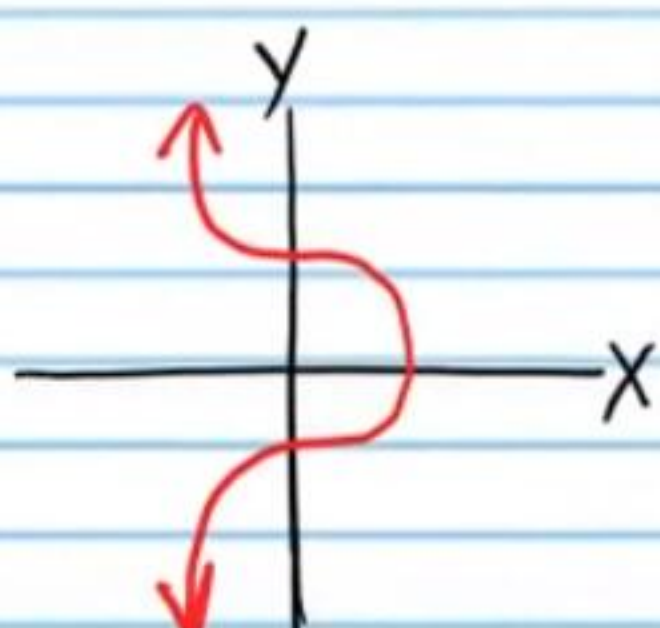
1-to-1 Function



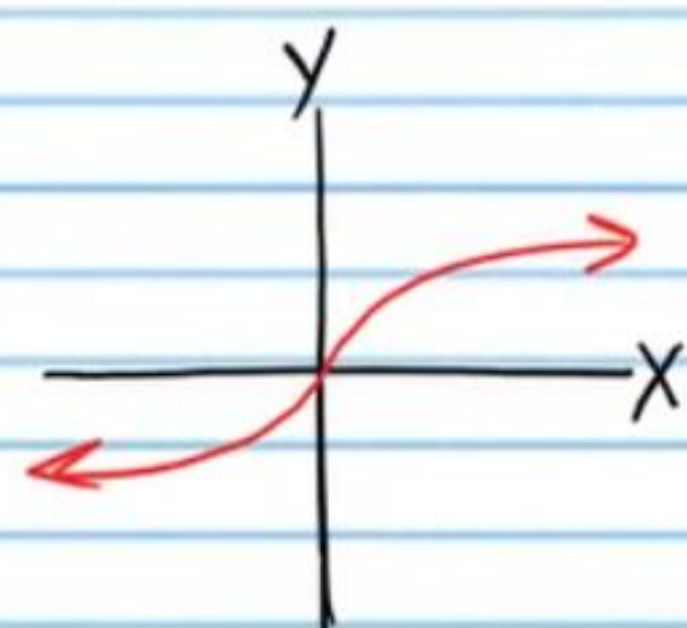
Not a Function



Function



Not a Function



1-to-1 Function

Inverse Functions

Inverse Function: A function that performs operations that are the reverse of another function in question.
 * Warning: The inverse of a function $f(x)$ is notated $f^{-1}(x)$, but -1 in this notation is not an exponent.

Two functions are inverses of each other if for every point $(x, f(x))$ on the graph of one function, there exists a point $(f(x), x)$ on the graph of the other function, assuming that each function has a unique output for every input (i.e. each function is "one-to-one"). If either function is not one-to-one, then restrictions must be applied to one or both of the functions so that they are perfectly matched inverses.

Each table below represents a function "f". Write the values of the inverse function "f⁻¹". If "f" is not a one-to-one function, then you must omit values of "f⁻¹" so that it qualifies as a function. To ensure that the functions are perfectly matched inverses, values of "f" may need to be omitted also. You may choose which values to omit. Highlight the omitted values.

(19)

X	f(x)
2	100
-4	36
0	-4
8	10
1	-7
-1	200

X	f ⁻¹ (x)
100	2
36	-4
-4	0
10	8
-7	1
200	-1

(20)

X	f(x)
17	20
18	21
19	25
20	31
21	42
22	80

X	f ⁻¹ (x)
20	17
21	18
25	19
31	20
42	21
80	22

21)

X	f(x)	X	f ⁻¹ (x)
5	-20	-20	5
-5	-21	-21	-5
6	-4	-4	6
-6	9	9	-6
7	6	6	7
9	6	6	9

22)

X	f(x)	X	f ⁻¹ (x)
4	7	7	4
8	10	10	8
14	20	20	14
20	10	10	20
30	70	70	30
40	10	10	40

23)

X	f(x)	X	f ⁻¹ (x)
2	1	1	2
3	4	4	3
8	7	7	8
5	-2	-2	5
10	3	3	10
11	99	99	11

24)

X	f(x)	X	f ⁻¹ (x)
-10	14	14	-10
-12	18	18	-12
-14	16	16	-14
-3	15	15	-3
2	9	9	2
1	8	8	1

25)

X	f(x)	X	f ⁻¹ (x)
1	33	33	1
2	34	34	2
3	34	34	3
4	35	35	4
5	36	36	5
6	37	37	6

26)

X	f(x)	X	f ⁻¹ (x)
2	8	8	2
-2	4	4	-2
1	8	8	1
17	8	8	17
3	8	8	3
4	8	8	4

If "f" and "g" are inverse functions, answer the following questions.

- 27) If $f(2) = 8$, find $g(8)$. $(8, g(8))$

$$f \rightarrow (2, 8) \rightarrow g \rightarrow (8, 2) \quad g(8) = 2$$

- 28) If $g(-3.1) = -7$, find $f(-7)$.

$$g \rightarrow (-3.1, -7) \rightarrow f \rightarrow (-7, -3.1) \quad f(-7) = -3.1$$

- 29) If $f(-\frac{4}{5}) = \frac{3}{7}$, find $g(\frac{3}{7})$.

$$f \rightarrow (-\frac{4}{5}, \frac{3}{7}) \rightarrow g \rightarrow (\frac{3}{7}, -\frac{4}{5}) \quad g(\frac{3}{7}) = -\frac{4}{5}$$

- 30) If $g(0) = -1$, find $f(-1)$.

$$g \rightarrow (0, -1) \rightarrow f \rightarrow (-1, 0) \quad f(-1) = 0$$

- 31) If $f(0.4) = -3\frac{1}{2}$, find $g(-3\frac{1}{2})$.

$$f \rightarrow (0.4, -3\frac{1}{2}) \rightarrow g \rightarrow (-3\frac{1}{2}, 0.4) \quad g(-3\frac{1}{2}) = 0.4$$

- 32) If $g(10) = 17$, find $f(17)$.

$$g \rightarrow (10, 17) \rightarrow f \rightarrow (17, 10) \quad f(17) = 10$$

- 33) If $f(-6) = -9$, find $g(-9)$.

$$f \rightarrow (-6, -9) \rightarrow g \rightarrow (-9, -6) \quad g(-9) = -6$$

- 34) If $g(-9.7) = \frac{8}{11}$, find $f(\frac{8}{11})$.

$$g \rightarrow (-9.7, \frac{8}{11}) \rightarrow f \rightarrow (\frac{8}{11}, -9.7) \quad f(\frac{8}{11}) = -9.7$$

In the following problems, "f" and "f⁻¹" are inverse functions.

• (35) If $f(2)=3$, find $f^{-1}(3)$.

$f \rightarrow (2, 3) \rightarrow f^{-1} \rightarrow (3, 2) \quad f^{-1}(3) = \boxed{2}$

(36) If $f(-4)=86$, find $f^{-1}(86)$.

$f \rightarrow (-4, 86) \rightarrow f^{-1} \rightarrow (86, -4) \quad f^{-1}(86) = \boxed{-4}$

(37) If $f^{-1}(18)=16$, find $f(16)$.

$f^{-1} \rightarrow (18, 16) \rightarrow f \rightarrow (16, 18) \quad f(16) = \boxed{18}$

(38) If $f^{-1}(\frac{3}{10})=-0.1$, find $f(-0.1)$.

$f^{-1} \rightarrow (\frac{3}{10}, -0.1) \rightarrow f \rightarrow (-0.1, \frac{3}{10}) \quad f(-0.1) = \boxed{\frac{3}{10}}$

• (39) If $f(-22)=17.1$, find $f^{-1}(17.1)$.

$f \rightarrow (-22, 17.1) \rightarrow f^{-1} \rightarrow (17.1, -22) \quad f^{-1}(17.1) = \boxed{-22}$

(40) If $f(0)=19$ find $f^{-1}(19)$.

$f \rightarrow (0, 19) \rightarrow f^{-1} \rightarrow (19, 0) \quad f^{-1}(19) = \boxed{0}$

(41) If $f^{-1}(-18)=-20$, find $f(-20)$.

$f^{-1} \rightarrow (-18, -20) \rightarrow f \rightarrow (-20, -18) \quad f(-20) = \boxed{-18}$

(42) If $f^{-1}(1.1)=-10$, find $f(-10)$.

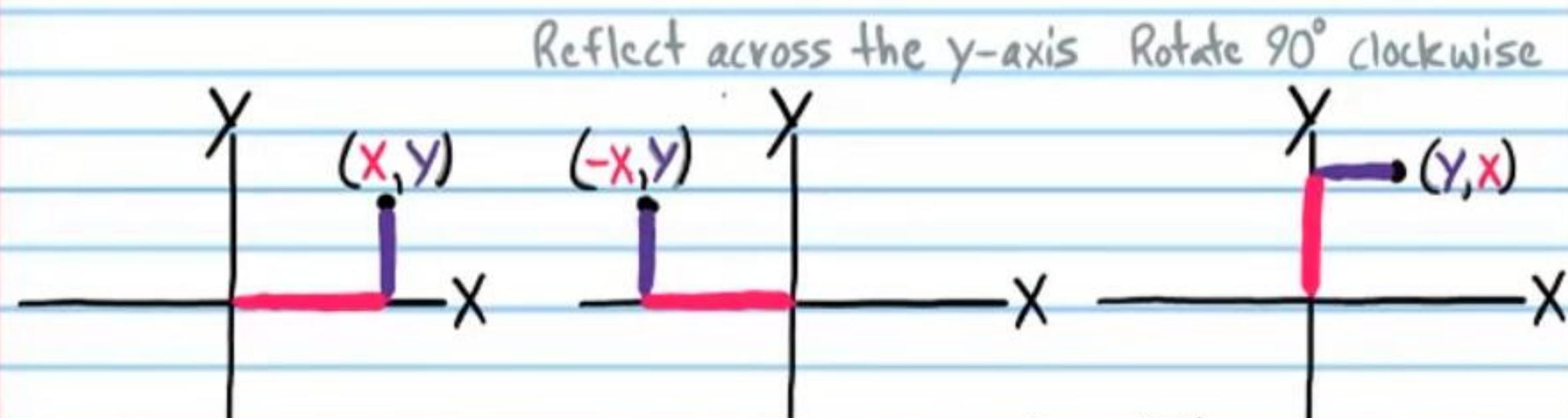
$f^{-1} \rightarrow (1.1, -10) \rightarrow f \rightarrow (-10, 1.1) \quad f(-10) = \boxed{1.1}$

Drawing the Graph of an Inverse Function

Original Graph

Step 1

Step 2



Each graph below represents a function "f." Draw the graph of the inverse function "f⁻¹." If "f" is not a one-to-one function, then you must erase parts of the graph of f⁻¹ so that it qualifies as a function. You may also highlight parts of the original graph that need to be erased so that the two inverses are perfectly matched. Use the Horizontal Line Test and the Vertical Line Test.

• (43)

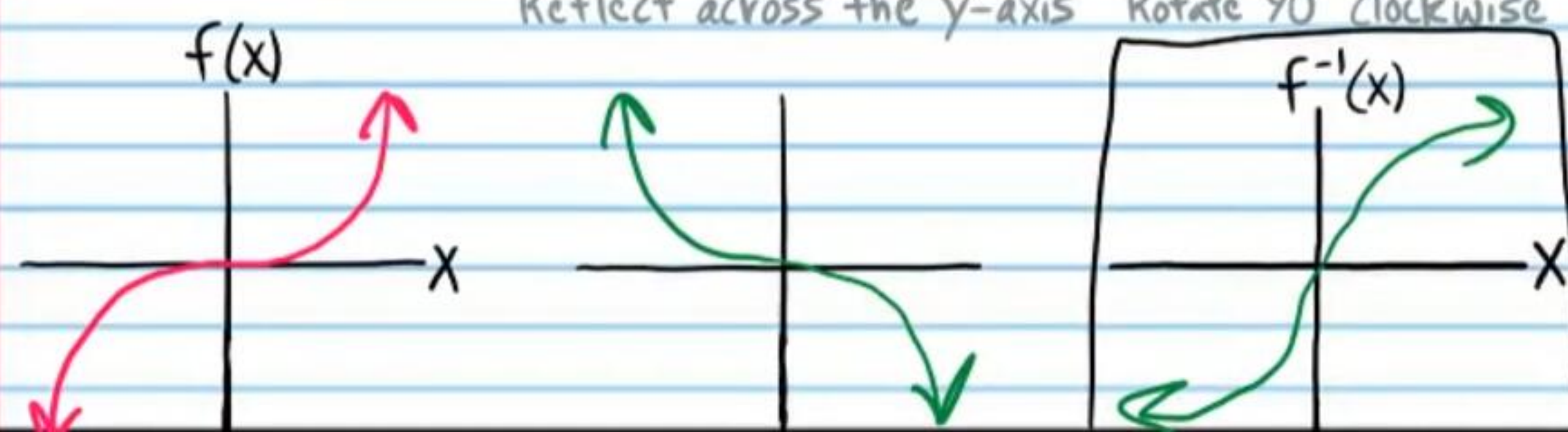
Write the inverse function
i.e. Reflection across $y=x$

Step 1

Reflect across the y-axis

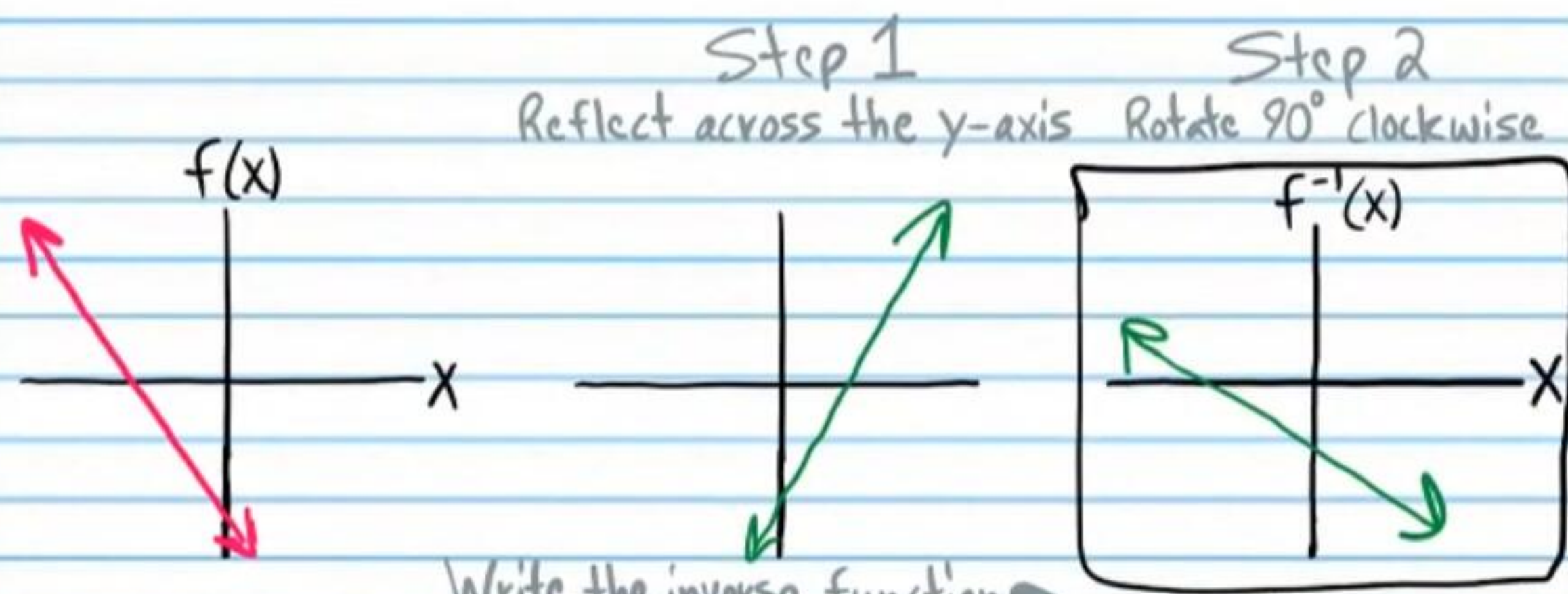
Step 2

Rotate 90° clockwise



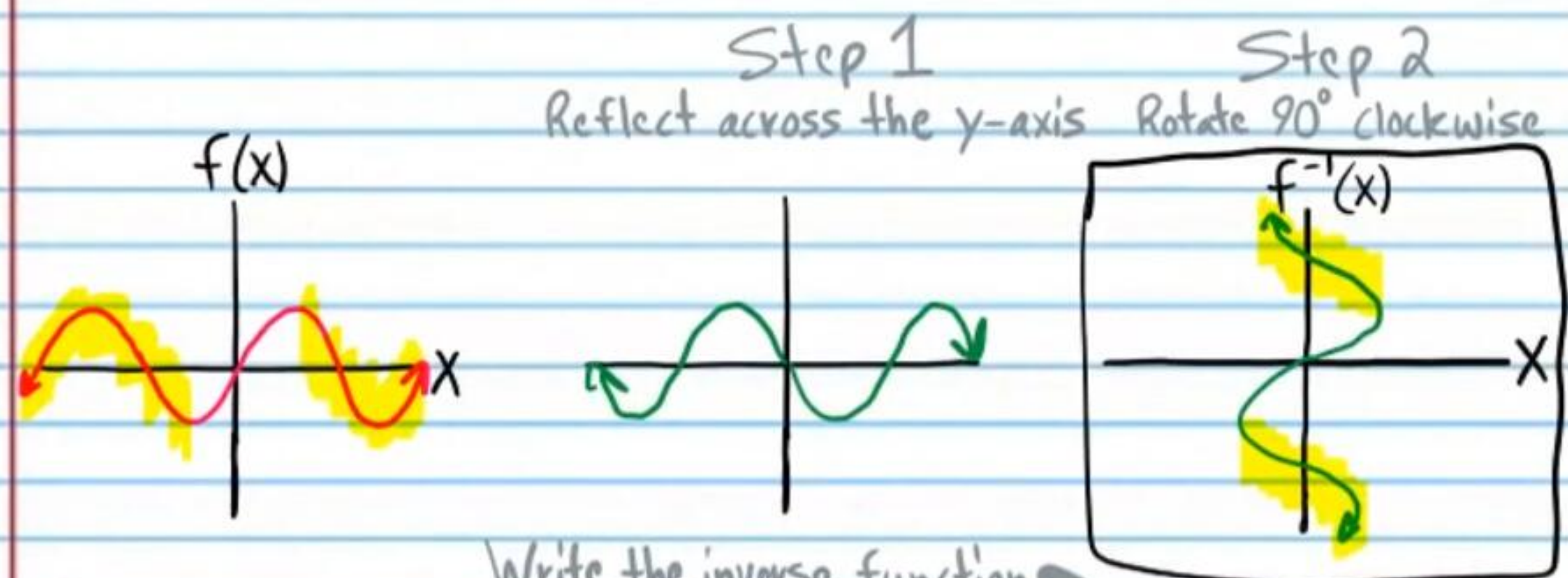
(44)

Write the inverse function
i.e. Reflection across $y=x$



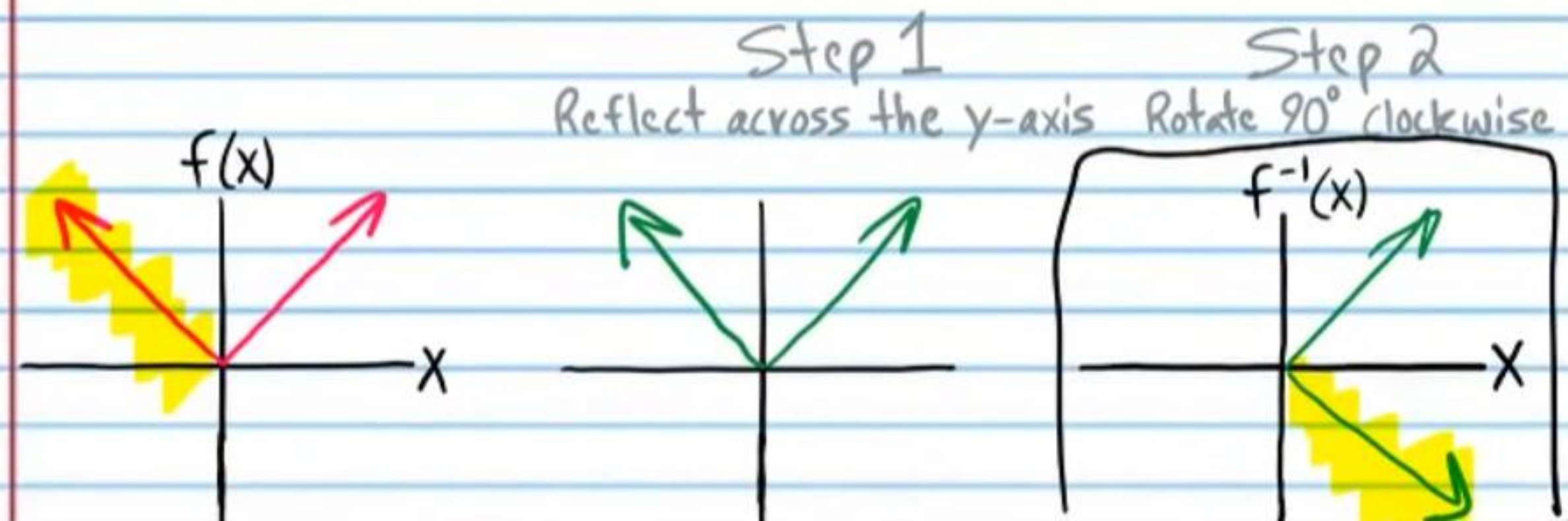
Write the inverse function
i.e. Reflection across $y=x$

(45)



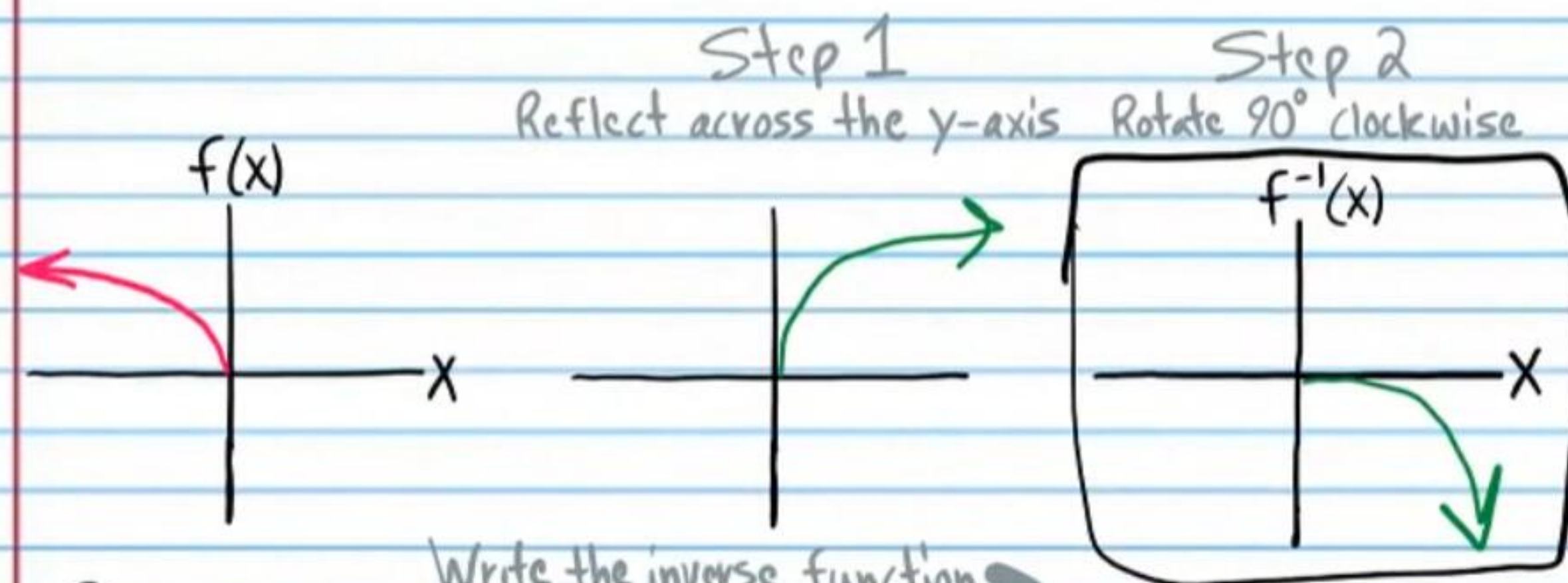
Write the inverse function
i.e. Reflection across $y=x$

(46)



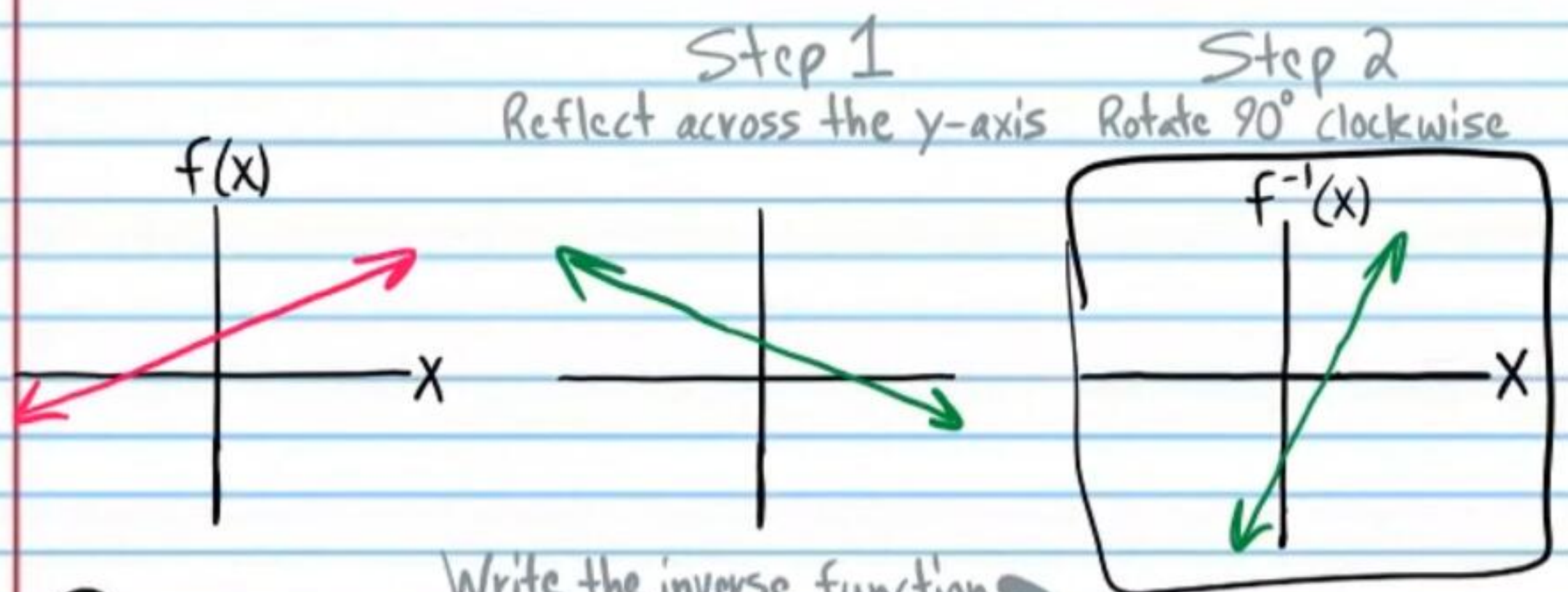
(47)

Write the inverse function
i.e. Reflection across $y=x$



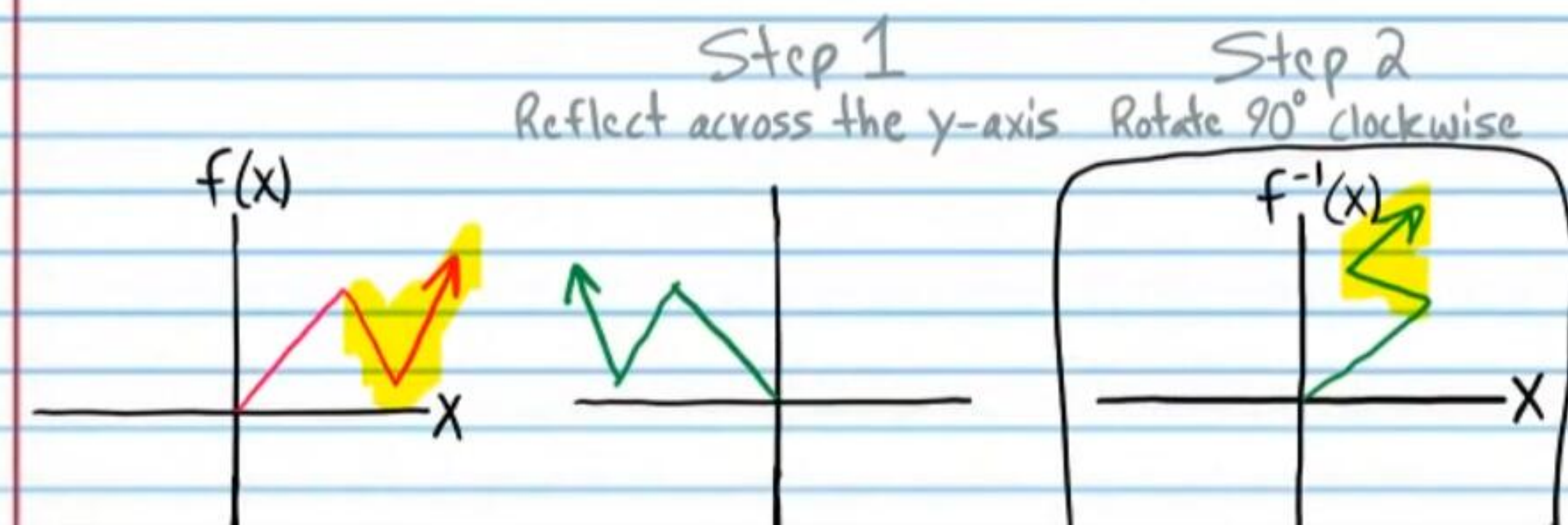
(48)

Write the inverse function
i.e. Reflection across $y=x$



(49)

Write the inverse function
i.e. Reflection across $y=x$

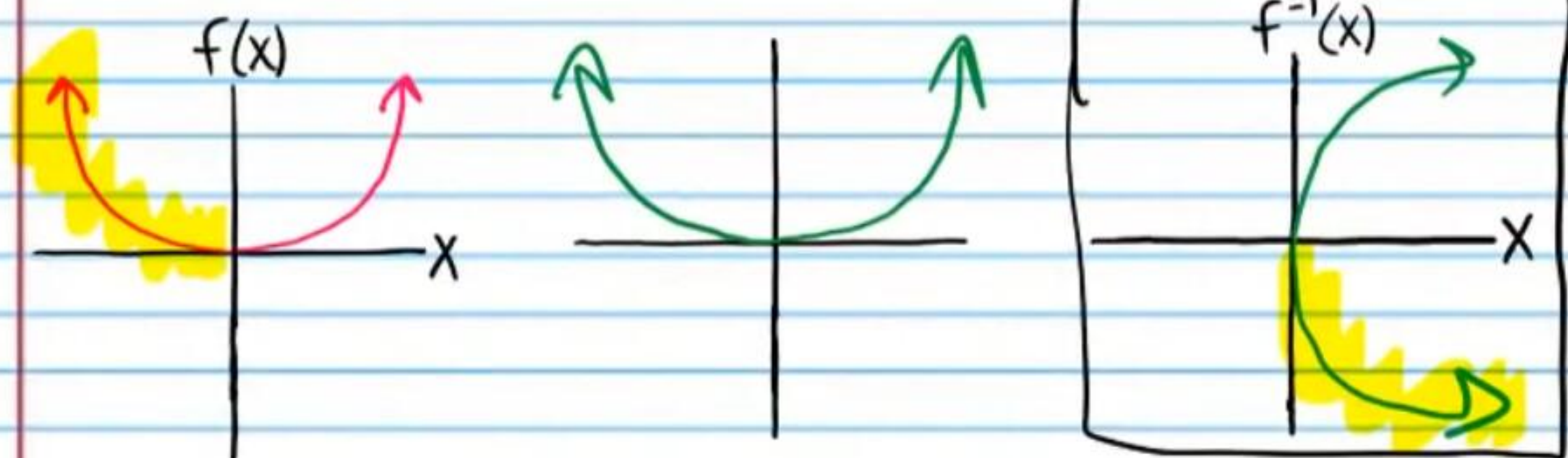


(50)

Write the inverse function
i.e. Reflection across $y=x$

Step 1
Reflect across the y -axis

Step 2
Rotate 90° clockwise



Finding the Equation of an Inverse Function

Example: Find the inverse of the function $f(x) = 2x + 3$.

Forward (i.e. the original function)

- i) Start x ii) Multiply by 2. $2x$ iii) Add 3. $2x + 3$

Reverse (i.e. the inverse function)

- iii) Divide by 2. $\frac{x-3}{2}$ ii) Subtract 3. $x-3$ i) Start x

Inverse Function: $f^{-1}(x) = \frac{x-3}{2}$

Steps for Finding the Equation of an Inverse Function

- I) Change $f(x)$ to y to make the equation easier to write.
- II) Replace x 's with y 's and y 's with x 's.
- III) Solve for y (i.e. isolate y).
- IV) Change the new isolated y to $f^{-1}(x)$.

Find the equation of the inverse function.

• 51) $f(x) = x^3 - 8$

I) $y = x^3 - 8$

II) $x = y^3 - 8$

III) $x + 8 = y^3$

$\sqrt[3]{x+8} = y$

IV) $\sqrt[3]{x+8} = f^{-1}(x)$

$f^{-1}(x) = \sqrt[3]{x+8}$

52) $f(x) = 6(x+10) + 7$

I) $y = 6(x+10) + 7$

II) $x = 6(y+10) + 7$

III) $x - 7 = 6(y+10)$

$x - 7 = 6y + 60$

$x - 67 = 6y$

$\frac{x-67}{6} = y$

IV) $f^{-1}(x) = \frac{x-67}{6}$

$$\textcircled{53} f(x) = \frac{x+2}{4}$$

$$\text{I) } y = \frac{x+2}{4}$$

$$\text{II) } x = \frac{y+2}{4}$$

$$\text{III) } (4)x = \frac{y+2}{4} (4)$$

$$4x = y+2$$

$$4x-2 = y$$

$$\text{IV) } 4x-2 = f^{-1}(x)$$

$$\boxed{f^{-1}(x) = 4x-2}$$

$$\textcircled{54} f(x) = 7-4x$$

$$\text{I) } y = 7-4x$$

$$\text{II) } x = 7-4y$$

$$\text{III) } x-7 = -4y$$

$$-\frac{x-7}{4} = y$$

$$\text{IV) } -\frac{x-7}{4} = f^{-1}(x)$$

$$\boxed{f^{-1}(x) = -\frac{x-7}{4}}$$

$$\bullet \textcircled{55} f(x) = \frac{x}{2} + 10$$

$$\text{I) } y = \frac{x}{2} + 10$$

$$\text{II) } x = \frac{y}{2} + 10$$

$$\text{III) } x-10 = \frac{y}{2}$$

$$2(x-10) = \frac{y}{2} (2)$$

$$2(x-10) = y$$

$$\text{IV) } 2(x-10) = f^{-1}(x)$$

$$\boxed{f^{-1}(x) = 2(x-10)}$$

$$\textcircled{56} f(x) = 2x^5 - 1$$

$$\text{I) } y = 2x^5 - 1$$

$$\text{II) } x = 2y^5 - 1$$

$$\text{III) } x+1 = 2y^5$$

$$\frac{x+1}{2} = y^5$$

$$\sqrt[5]{\frac{x+1}{2}} = y$$

$$\text{IV) } \sqrt[5]{\frac{x+1}{2}} = f^{-1}(x)$$

$$\boxed{f^{-1}(x) = \sqrt[5]{\frac{x+1}{2}}}$$

$$\textcircled{57} f(x) = -2(x-4)+1$$

$$\text{I) } y = -2(x-4)+1$$

$$\text{II) } x = -2(y-4)+1$$

$$\text{III) } x-1 = -2(y-4)$$

$$\frac{x-1}{-2} = \frac{-2(y-4)}{-2}$$

$$-\frac{x-1}{2} = y-4$$

$$-\frac{x-1}{2} + 4 = y$$

$$\text{IV) } -\frac{x-1}{2} + 4 = f^{-1}(x)$$

$$\boxed{f^{-1}(x) = -\frac{x-1}{2} + 4}$$

$$\textcircled{58} f(x) = 2x-23$$

$$\text{I) } y = 2x-23$$

$$\text{II) } x = 2y-23$$

$$\text{III) } x+23 = 2y$$

$$\frac{x+23}{2} = y$$

$$\text{IV) } \frac{x+23}{2} = f^{-1}(x)$$

$$\boxed{f^{-1}(x) = \frac{x+23}{2}}$$

$$\bullet \textcircled{59} f(x) = 8^x$$

$$\text{I) } y = 8^x$$

$$\text{II) } x = 8^y$$

$$\text{III) } y = \log_8 x$$

$$\text{IV) } \boxed{f^{-1}(x) = \log_8 x}$$

$$\bullet \textcircled{61} f(x) = 5^x$$

$$\text{I) } y = 5^x$$

$$\text{II) } x = 5^y$$

$$\text{III) } y = \log_5 x$$

$$\text{IV) } \boxed{f^{-1}(x) = \log_5 x}$$

$$\textcircled{60} f(x) = 7^x$$

$$\text{I) } y = 7^x$$

$$\text{II) } x = 7^y$$

$$\text{III) } y = \log_7 x$$

$$\text{IV) } \boxed{f^{-1}(x) = \log_7 x}$$

$$\textcircled{62} f(x) = 11^x$$

$$\text{I) } y = 11^x$$

$$\text{II) } x = 11^y$$

$$\text{III) } y = \log_{11} x$$

$$\text{IV) } \boxed{f^{-1}(x) = \log_{11} x}$$

⑥③ $f(x) = \log_9 x$

I) $y = \log_9 x$

II) $x = \log_9 y$

III) $y = 9^x$

IV) $f^{-1}(x) = 9^x$

⑥⑤ $f(x) = \log_6 x$

I) $y = \log_6 x$

II) $x = \log_6 y$

III) $y = 6^x$

IV) $f^{-1}(x) = 6^x$

⑥④ $f(x) = \log_{18} x$

I) $y = \log_{18} x$

II) $x = \log_{18} y$

III) $y = 18^x$

IV) $f^{-1}(x) = 18^x$

⑥⑥ $f(x) = \log_2 x$

I) $y = \log_2 x$

II) $x = \log_2 y$

III) $y = 2^x$

IV) $f^{-1}(x) = 2^x$

In the following problems, restrictions must be used on the original function and/or inverse function so that they are perfectly matched and no aberrations occur. To understand this, graph the function and note that it fails the Horizontal Line Test, which means that it is not one-to-one. Draw the inverse graph and note that it fails the Vertical Line Test. Highlight the parts of the inverse graph and the original graph that you will omit and use inequality notation next to the equations to indicate restrictions.

⑥⑦ $f(x) = (x-2)^2$, $x \geq 2$

I) $y = (x-2)^2$

II) $x = (y-2)^2$

III) $\pm\sqrt{x} = \sqrt{(y-2)^2}$

$\pm\sqrt{x} = y-2$

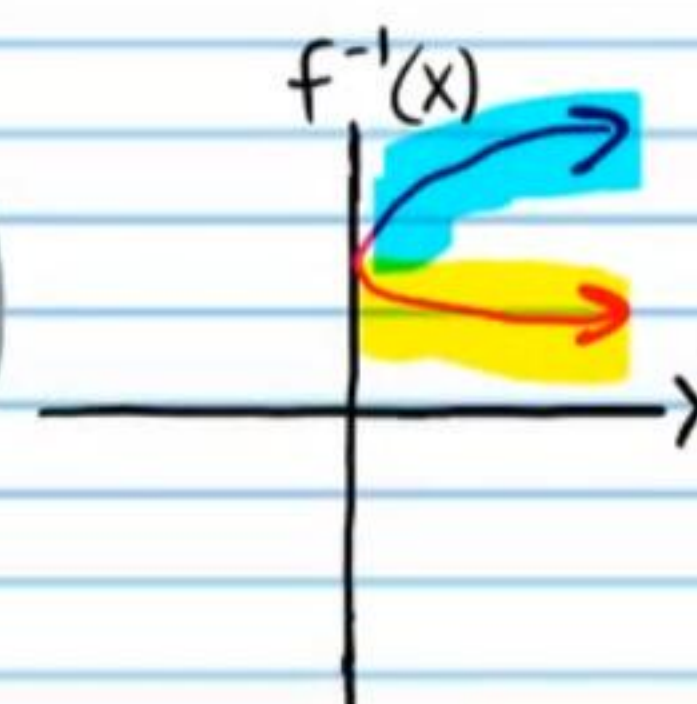
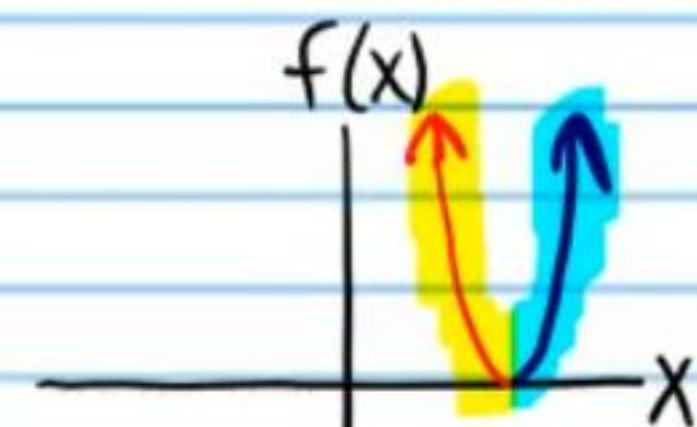
$\pm\sqrt{x} + 2 = y$

IV) $\pm\sqrt{x} + 2 = f^{-1}(x)$

$f^{-1}(x) = \sqrt{x} + 2$

Reflect across the y-axis.

Rotate 90° clockwise.



⑥⑧ $f(x) = \sqrt{x} + 5$

I) $y = \sqrt{x} + 5$

II) $x = \sqrt{y} + 5$

III) $x - 5 = \sqrt{y}$

$$(x-5)^2 = (\sqrt{y})^2$$

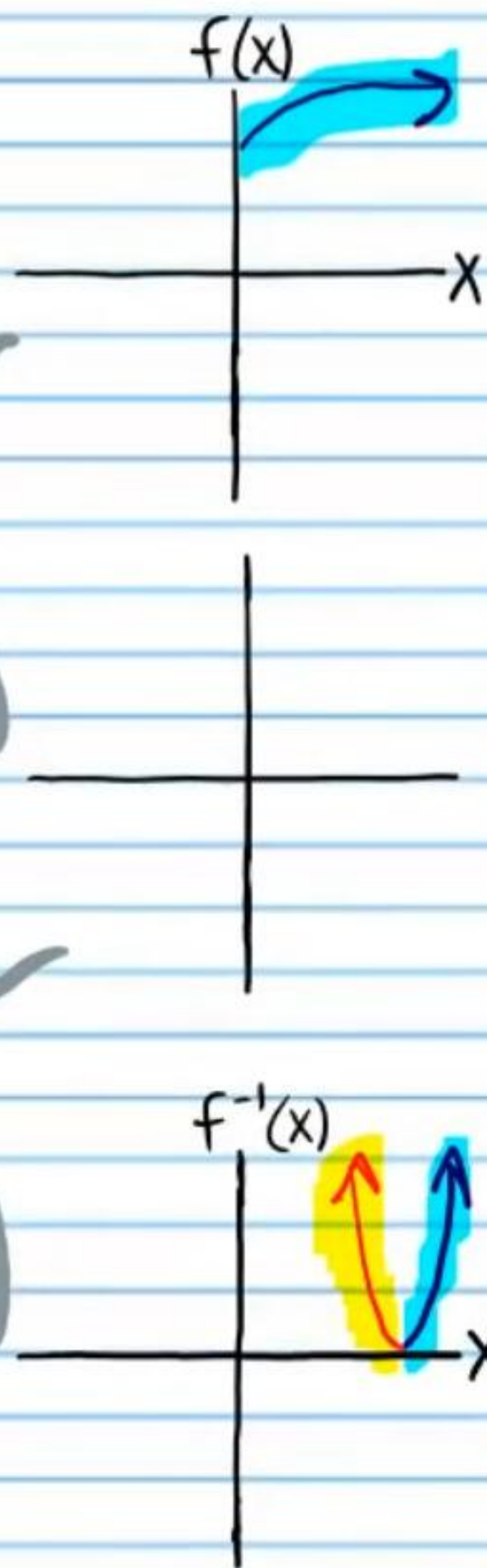
$$(x-5)^2 = y$$

IV) $(x-5)^2 = f^{-1}(x)$

$$f^{-1}(x) = (x-5)^2, \quad x \geq 5$$

Reflect
across
the
y-axis.

Rotate
90°
clockwise.



• ⑥⑨ $f(x) = x^2 - 1, \quad x \geq 0$

I) $y = x^2 - 1$

II) $x = y^2 - 1$

III) $x + 1 = y^2$

$$\pm \sqrt{x+1} = \sqrt{y^2}$$

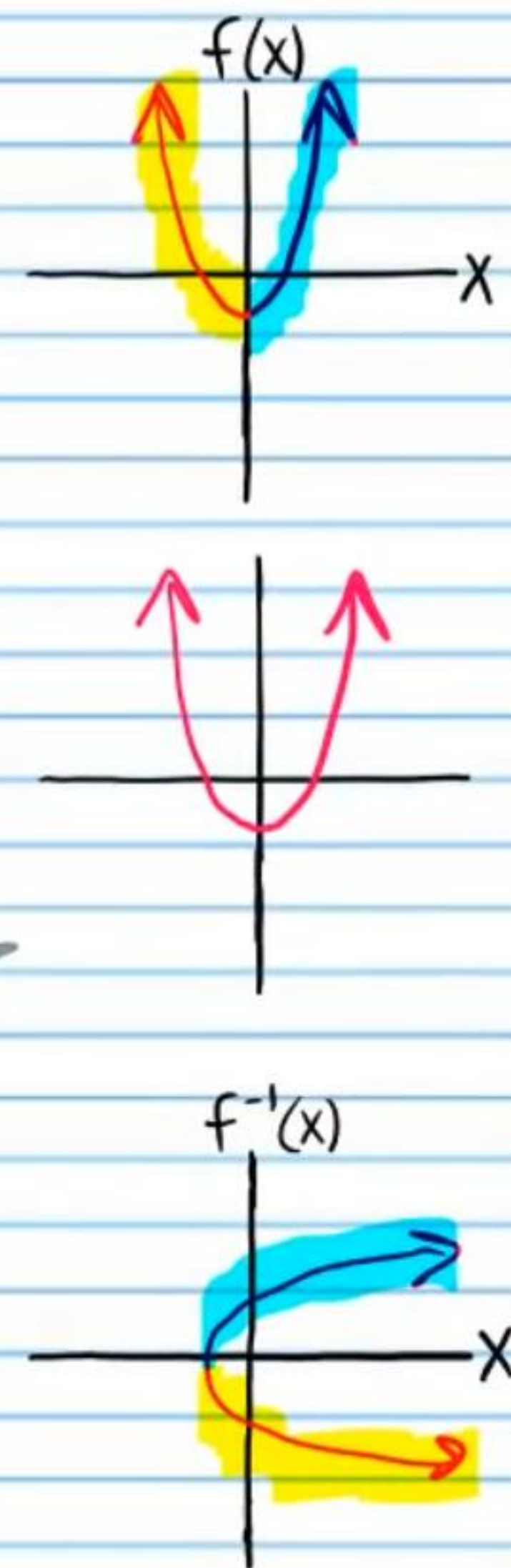
$$\pm \sqrt{x+1} = y$$

IV) $\pm \sqrt{x+1} \neq f^{-1}(x)$

$$f^{-1}(x) = \sqrt{x+1}$$

Reflect
across
the
y-axis.

Rotate
90°
clockwise.



70) $f(x) = \sqrt{x-3}$

I) $y = \sqrt{x-3}$

II) $x = \sqrt{y-3}$

III) $x^2 = (\sqrt{y-3})^2$

$x^2 = y-3$

$x^2 + 3 = y$

IV) $x^2 + 3 = f^{-1}(x)$

$f^{-1}(x) = x^2 + 3, x \geq 0$

